Comparison between species

Once the topic was defined, we began a more in-depth investigation of the brain characteristics of each species.

We started by analyzing the human brain as a reference point, along with those of bees, ants, elephants, octopuses, fish, birds, mice, and pigs. As mentioned previously, our main analytical criteria were evolution, environment, intelligence, and emotions.

We aimed to gather as much information as possible in order to determine which species would be the most compelling to contrast, using humans as the reference.

This stage resulted in an extensive body of information and led us to realize that our next steps needed to focus on synthesizing the data and deciding which species were most worthwhile to compare within the framework of our project.

Human
- Social intelligence
- Problem-solving intelligence / tool use
- Communicative intelligence (language)
- Memory / abstraction
- Self-awareness
- Sensory and motor intelligence (diverse)

Bee
- Swarm intelligence
- Communicative intelligence (dance language)
- Memory abilities (flower recognition, navigation)
- Sensory intelligence (recognizing patterns, colors, scents)

Ant
- Swarm intelligence
- Basic problem-solving
- Communicative intelligence (pheromone trails, body signals)
- Sensory intelligence (chemical navigation, cooperation)

Elephant
- Social intelligence
- Problem-solving intelligence (tool use, cooperation)
- Communicative intelligence (body language, infrasound)
- Memory / abstraction (very strong)
- Self-awareness

Octopus / Squid
- Problem-solving intelligence (tools & puzzles)
- Memory (spatial, tactical)
- Signs of self-awareness (exploratory behavior)
- Sensory intelligence (color and structure recognition)

Fish
- Swarm intelligence
- Problem-solving (basic, for foraging)
- Learning ability / memory (places, patterns)
- Communicative intelligence (body language, acoustic signals in some species)

Bird
- Problem-solving intelligence (tool use, e.g. crows)
- Social intelligence (parrots, corvids)
- Memory / abstraction (e.g. jays)
- Communicative intelligence (song and call repertoires)
- Signs of self-awareness (magpies, corvids)

Mouse
- Social intelligence (group behavior)
- Problem-solving (mazes, foraging)
- Learning ability / memory (locations, stimuli)
- Sensory intelligence (smell, tactile senses)

Pig
- Social intelligence
- Problem-solving intelligence (foraging, games)
- Memory / abstraction
- Self-awareness (mirror test passed)
- Communicative intelligence (sounds, gestures)

This overview enables a quick assignment of intelligence forms to the listed animals, based on current research.

Basic ant brain biology
Ant brains are extremely small (a fraction of a milligram) but contain several hundred thousand neurons organized into distinct regions. Key structures include optic lobes for vision, antennal lobes for smell, and mushroom bodies for learning and memory, which are especially important for complex behavior. Ants use many of the same neurotransmitters as vertebrates, such as dopamine, serotonin, and octopamine, which support motivation, learning, and internal states.
- The antennal lobes process chemical signals (pheromones) and are enlarged in highly social species that rely heavily on smell for communication.
- The mushroom bodies expand in species and castes that need advanced navigation and learning, such as foragers and fungus-growing ants.
- Brain investment (which regions are large or small) shifts with colony size and division of labor, indicating strong evolutionary tuning between brain and social life.

Habitat and environment
Ants live in a wide range of habitats, from tropical rainforests to deserts, and their brains reflect the specific sensory and cognitive demands of these environments. Species navigating open, visually complex landscapes invest more in visual processing and spatial memory, while those in dark soil or wood rely more on chemical and tactile information

Ant brains are tiny but highly specialized organs that evolved to control life in dense social colonies and challenging environments, leading to sophisticated navigation, communication, and collective "intelligence" without human-like consciousness or emotions. Their structure, behavior, and possible emotion-like states all make sense only in relation to the ecological and social world in which ants live.
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- Desert ants use landmark learning, path integration, optic flow, and celestial cues to find their way, requiring accurate spatial memory and integration of multiple sensory inputs in the brain.
- Fungus-growing and leaf-cutting ants must coordinate farming, foraging, and nest maintenance, which selects for brain circuits that handle complex odor discrimination and learned routes.
- Because colonies are often crowded and multi-layered, ant brains emphasize efficient processing of local cues rather than building detailed "world models" like those of mammals.

This environmental coupling explains why some ant species have enlarged mushroom bodies (for flexible navigation and learning).

Social life and brain organization
Ant colonies are eusocial "super-organisms" where thousands or millions of individuals divide labor between queens, workers, and males, and among worker roles such as nurses and foragers.